

Measures of central tendency

Mean, median, and mode are the measures of central tendency, used to study the various characteristics of a given set of data. A measure of central tendency describes a set of data by identifying the central position in the data set as a single value. We can think of it as a tendency of data to cluster around a middle value. In statistics, the three most common measures of central tendencies are mean, median and mode.

➤ **Mean:**

The mean is also known as the average, and it is calculated by adding up all the values in a data set and dividing by the total number of values.

➤ **Median:**

The median is the middle value of a data set, which separates the highest and lowest values equally. It is calculated by arranging the data set in order from lowest to highest and finding the value in the exact middle.

➤ **Mode:**

The mode is the value that appears most frequently in a data set.

Mean

Choosing the best measures of central tendency depends on the type of data we have. Let's begin by understanding the meaning of each of these terms.

The arithmetic mean of a given data is the sum of all observations divided by the number of observations. For example, a cricketer's scores in five ODI matches are as follows: 12, 34, 45, 50, 24. To find his average score in a match, we calculate the arithmetic mean of data using the mean formula:

Mean = Sum of all observations / Number of observations

$$\Rightarrow \text{Mean} = (12 + 34 + 45 + 50 + 24) / 5$$

$$\Rightarrow \text{Mean} = 165 / 5$$

$$= 33$$

Mean is denoted by $\bar{x}$ (pronounced as x bar). Mean is calculated a little differently when the data is grouped or ungrouped. Let's find the mean in both cases.

Types of Mean

There are majorly three different types of mean value that you will be studying in statistics.

1. Arithmetic Mean
2. Geometric Mean
3. Harmonic Mean

Arithmetic Mean

When you add up all the values and divide by the number of values it is called Arithmetic Mean. To calculate, just add up all the given numbers then divide by how many numbers are given.

Example: What is the mean of 3, 5, 9, 5, 7, 2?

Now add up all the given numbers:

$$3 + 5 + 9 + 5 + 7 + 2 = 31$$

Now divide by how many numbers are provided in the sequence:

$$\frac{31}{6} = 5.16$$

5.16 is the answer.

Geometric Mean

The geometric mean of two numbers x and y is $\sqrt{xy}$. If you have three numbers x , y , and z , their geometric mean is $\sqrt[3]{xyz}$.

$$\text{Geometric mean} = \sqrt[n]{x_1 x_2 x_3 \dots \dots \dots x_n} = (x_1 x_2 x_3 \dots \dots \dots x_n)^{\frac{1}{n}}$$

Example: Find the geometric mean of 4 and 3?

$$\begin{aligned} \text{Geometric mean} &= \sqrt{4 \times 3} \\ &= 3.46 \end{aligned}$$

Harmonic Mean

The harmonic mean is used to average ratios. For two numbers x and y , the harmonic mean is $\frac{2xy}{(x+y)}$. For, three numbers x , y , and z , the harmonic mean is $\frac{3xyz}{(xy+xz+yz)}$.

$$\text{Harmonic mean} = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} \dots \dots \dots \frac{1}{x_n}}$$

Root Mean Square (Quadratic)

The root mean square is used in many engineering and statistical applications, especially when there are data points that can be negative.

$$X_{rms} = \sqrt{\frac{x_1^2 + x_2^2 + x_3^2 + \dots \dots \dots + x_n^2}{n}}$$

Contraharmonic Mean

The contraharmonic mean of x and y is $(x^2 + y^2)/(x + y)$. For n values,

$$= \frac{x_1^2 + x_2^2 + x_3^2 + \dots \dots \dots + x_n^2}{x_1 + x_2 + x_3 + \dots \dots \dots + x_n}$$

Mean of Ungrouped Data

Let $x_1, x_2, x_3, \dots, x_n$ be n observations. We can find the arithmetic mean using the mean formula:

$$\text{Mean, } \bar{x} = \frac{(x_1 + x_2 + \dots + x_n)}{n}$$

Example: If the heights of 5 people are 142 cm, 150 cm, 149 cm, 156 cm, and 153 cm. Find the mean height.

$$\begin{aligned}\text{Mean height, } \bar{x} &= \frac{(142 + 150 + 149 + 156 + 153)}{5} \\ &= \frac{750}{5} \\ &= 150\end{aligned}$$

Mean, $\bar{x} = 150 \text{ cm}$

Thus, the mean height is 150 cm.

Mean of Grouped Data

When the data is present in tabular form, we use the following formula:

$$\text{Mean, } \bar{x} = \frac{(x_1f_1 + x_2f_2 + \dots + x_nf_n)}{(f_1 + f_2 + \dots + f_n)}$$

Example 1: Find the mean of the following distribution:

x	4	6	9	10	15
f	5	10	10	7	8

Solution:

Calculation table for arithmetic mean:

x_i	f_i	$x_i f_i$
4	5	20
6	10	60
9	10	90
10	7	70
15	8	120
	$\sum f_i = 40$	$\sum x_i f_i = 360$

$$\begin{aligned}\text{Mean, } \bar{x} &= \frac{\sum x_i f_i}{\sum f_i} \\ &= \frac{360}{40} \\ &= 9\end{aligned}$$

Thus, $\text{Mean} = 9$

Example 2: Here is an example where the data is in the form of class intervals. The following table indicates the data on the number of patients visiting a hospital in a month. Find the average number of patients visiting the hospital in a day.

Number of patients	Number of days visiting hospital
0-10	2
10-20	6
20-30	9
30-40	7
40-50	4
50-60	2

Solution: In this case, we find the class mark (also called as mid-point of a class) for each class.

Note: $\text{Class mark} = \frac{(\text{lower limit} + \text{upper limit})}{2}$

Let $x_1, x_2, x_3, \dots, x_n$ be the class marks of the respective classes.

Hence, we get the following table:

Class	Class mark (x_i)	frequency (f_i)	$x_i f_i$
0-10	5	2	10
10-20	15	6	90
20-30	25	9	225
30-40	35	7	245
40-50	45	4	180
50-60	55	2	110
Total:		$\sum f_i = 30$	$\sum x_i f_i = 860$

$$\begin{aligned}
 \text{Mean, } \bar{x} &= \frac{\sum x_i f_i}{\sum f_i} \\
 &= \frac{860}{30} \\
 &= 28.67. \text{ Thus, Mean} = 28.67
 \end{aligned}$$

Example 3: Computation of arithmetic mean for the wage data of Beximco Pharmaceutical Company

Weekly wage in TK	48.5 - 53.5	53.5 - 58.5	58.5 - 63.5	63.5 - 68.5	68.5 - 73.5	73.5 - 78.5	78.5 - 83.5	83.5 - 88.5	88.5 - 93.5	93.5 - 98.5	Total
Frequency	2	2	3	5	5	5	5	7	10	6	50

Solution:

Calculation table:

Weekly wage in TK	Mid-point(z_i)	Frequency(f_i)	$f_i z_i$
48.5-53.5	51	2	102
53.5-58.5	56	2	112
58.5-63.5	61	3	183
63.5-68.5	66	5	330
68.5-73.5	71	5	355
73.5-78.5	76	5	380
78.5-83.5	81	5	405
83.5-88.5	86	7	602
88.5-93.5	91	10	910
93.5-98.5	96	6	576
Total		50	3955

$$\begin{aligned}\text{Mean, } \bar{x} &= \frac{\sum z_i f_i}{\sum f_i} \\ &= \frac{3955}{50} \\ &= 79.1.\end{aligned}$$

Thus, $\text{Mean} = 79.1$

Short-cut Method:

$$\bar{x} = a + \bar{u}h$$

Where,

a = Shifted origin/assumed mean

h = Class width

$\bar{u}$ = Mean of class deviation

$$\bar{u} = \frac{\sum u_i f_i}{\sum f_i}, \quad \left(\text{here, } u_i = \frac{x_i - a}{h} \text{ or } u_i = \frac{z_i - a}{h} \right)$$

Example 4: Compute mean of data from example 3 by using Short-cut Method.

Solution:

Weekly wage in TK	Mid-point(z_i)	Frequency(f_i)	$u_i = \frac{z_i - a}{h}$	$f_i u_i$
48.5-53.5	51	2	-4	-8
53.5-58.5	56	2	-3	-6
58.5-63.5	61	3	-2	-6
63.5-68.5	66	5	-1	-5
68.5-73.5	71 $\Rightarrow a$	5	0	0
73.5-78.5	76	5	1	5
78.5-83.5	81	5	2	10
83.5-88.5	86	7	3	21
88.5-93.5	91	10	4	40
93.5-98.5	96	6	5	30
Total		50		81

Here, $a = 71$ and $h = 5$

Thus, $\bar{u} = \frac{\sum u_i f_i}{\sum f_i}$

$$\Rightarrow \bar{u} = \frac{81}{50} = 1.62$$

Hence,

$$\begin{aligned}\bar{x} &= a + \bar{u}h \\ \Rightarrow \bar{x} &= 71 + 1.62 \times 5 \\ \therefore \bar{x} &= 79.1\end{aligned}$$

Theorem:

For a set of n non-zero positive values $x_1, x_2, \dots, x_n$. Prove that $A \geq G \geq H$, where A , G and H are respectively arithmetic mean, geometric mean and harmonic mean.

Proof:

By the definition

$$A = \frac{\sum_{i=1}^n x_i}{n}$$

and

$$G = (x_1 x_2 x_3 \dots x_n)^{\frac{1}{n}}$$

Taking logarithm of G ,

$$\log G = \log(x_1 x_2 x_3 \dots x_n)^{\frac{1}{n}}$$

$$\begin{aligned}
\Rightarrow \log G &= \frac{1}{n} \log(x_1 x_2 x_3 \dots x_n) \\
\Rightarrow \log G &= \frac{1}{n} \{ \log x_1 + \log x_2 + \log x_3 + \dots + \log x_n \} \\
\Rightarrow \log G &= \frac{1}{n} \sum_{i=1}^n \log x_i \\
\Rightarrow \log G &= \frac{1}{n} \sum_{i=1}^n \log(A - A + x_i) \\
\Rightarrow \log G &= \frac{1}{n} \sum_{i=1}^n \log A \left(1 - \frac{A - x_i}{A}\right) \\
\Rightarrow \log G &= \frac{1}{n} \sum_{i=1}^n \log A + \frac{1}{n} \sum_{i=1}^n \log \left(1 - \frac{A - x_i}{A}\right) \\
\Rightarrow \log G &= \log A + \frac{1}{n} \sum_{i=1}^n \left\{ -\frac{A - x_i}{A} - \frac{1}{2} \left(\frac{A - x_i}{A}\right)^2 - \frac{1}{3} \left(\frac{A - x_i}{A}\right)^3 \dots \right\} \quad ; [\text{note: } \log(1 - x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \dots] \\
\Rightarrow \log G &= \log A - \frac{1}{n} \sum_{i=1}^n \left\{ \frac{A - x_i}{A} + \frac{1}{2} \left(\frac{A - x_i}{A}\right)^2 + \frac{1}{3} \left(\frac{A - x_i}{A}\right)^3 \dots \right\} \\
\Rightarrow \log G &= \log A - \text{a positive number} \\
\Rightarrow \log G &\leq \log A \\
\text{Hence } A &\geq G \quad (1)
\end{aligned}$$

Using the relation $A \geq G$, we have

$$\begin{aligned}
\frac{x_1 + x_2 + \dots + x_n}{n} &\geq (x_1 x_2 x_3 \dots x_n)^{\frac{1}{n}} \\
\text{Replacing } x_1, x_2, \dots, x_n \text{ by } \frac{1}{x_1}, \frac{1}{x_2}, \frac{1}{x_3} \dots \frac{1}{x_n} \text{ respectively} \\
\frac{\frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} \dots \frac{1}{x_n}}{n} &\geq \left(\frac{1}{x_1} \cdot \frac{1}{x_2} \cdot \frac{1}{x_3} \dots \frac{1}{x_n} \right)^{\frac{1}{n}} \\
&\Rightarrow \frac{1}{H} \geq \frac{1}{G} \\
&\therefore G \geq H \quad (2)
\end{aligned}$$

Combining (1) and (2) we have,

$$A \geq G \geq H$$

Hence proved.

Theorem:

A variable x takes on n values which are in geometric progression. Obtain the arithmetic mean (AM), geometric mean (GM) and harmonic mean (HM) and hence show that

$$AM \times HM = GM^2$$

Proof:

Let the variable assume values $a, ar, ar^2, \dots, ar^{n-1}$ which are in geometric progression. Then by definition,

$$\begin{aligned}
AM &= \frac{a + ar + ar^2 + \dots + ar^{n-1}}{n} \\
&= \frac{a(1-r^n)}{n(1-r)}; \quad r < 1
\end{aligned}$$

$$GM = (a \cdot ar \cdot ar^2 \dots ar^{n-1})^{\frac{1}{n}}$$

$$\Rightarrow GM = (a^n \cdot r^{1+2+\dots+(n-1)})^{\frac{1}{n}}$$

$$\Rightarrow GM = a(r^{\frac{n(n-1)}{2}})^{\frac{1}{n}}$$

$$\Rightarrow GM = ar^{\frac{n-1}{2}}$$

$$\therefore GM^2 = a^2 r^{n-1}$$

Again,

$$\begin{aligned} HM &= \frac{n}{\frac{1}{a} + \frac{1}{ar} + \frac{1}{ar^2} + \dots + \frac{1}{ar^{n-1}}} \\ &= \frac{na}{1 + \frac{1}{r} + \frac{1}{r^2} + \dots + \frac{1}{r^{n-1}}} \\ &= \frac{na}{\frac{(\frac{1}{r})^{n-1}}{\frac{1}{r}-1}}; \quad [S = \frac{a(r^n-1)}{r-1}] \\ &= \frac{na}{\frac{1}{r^n} - 1} \\ &= \frac{na}{\frac{1}{r} - 1} \\ &= \frac{na}{\frac{1-r^n}{r^n}} \\ &= \frac{na}{\frac{1-r^n}{r}} \\ &= \frac{nar^n(1-r)}{r(1-r^n)} \\ HM &= \frac{nar^{n-1}(1-r)}{(1-r^n)} \end{aligned}$$

$$\text{Now, } AM \times HM = \frac{a(1-r^n)}{n(1-r)} \times \frac{nar^{n-1}(1-r)}{(1-r^n)}$$

$$\Rightarrow AM \times HM = a^2 r^{n-1}$$

$$\therefore AM \times HM = GM^2$$

Hence proved.

Real-life Applications of Mean

In the real world, when there is huge data available, we use statistics to deal with it. Suppose, in a data table, the price values of 10 clothing materials are mentioned. If we have to find the mean of the prices, then add the prices of each clothing material and divide the total sum by 10. It will result in an average value. Another example is that if we have to find the average age of students of a class, we have to add the age of individual students present in the class and then divide the sum by the total number of students present in the class.

Median

The value of the middlemost observation, obtained after arranging the data in ascending or descending order, is called the median of the data. For example, consider the data: 4, 4, 6, 3, 2. Let's arrange this data in ascending order: 2, 3, 4, 4, 6. There are 5 observations. Thus, *median = middle value* i.e. 4.

Median of Ungrouped Data

Step 1: Arrange the data in ascending or descending order.

Step 2: Let the total number of observations be n .

To find the median, we need to consider if n is even or odd.

If n is odd, then use the formula:

$$\text{Median} = \left[\frac{(n+1)}{2} \right] \text{th observation}$$

Example 1: Let's consider the data: 56, 67, 54, 34, 78, 43, 23. What is the median?

Solution: Arranging in ascending order, we get: 23, 34, 43, 54, 56, 67, 78.

Here, n (number of observations) = 7

$$\text{So, } \left[\frac{(7+1)}{2} \right] = 4$$

$\therefore$ Median = 4th observation

Median = 54

If n is even, then use the formula:

$$\text{Median} = \frac{\left[\frac{n}{2} \text{th obs.} + \left(\frac{n}{2} + 1 \right) \text{th obs.} \right]}{2}$$

Example 2: Let's consider the data: 50, 67, 24, 34, 78, 43. What is the median?

Solution: Arranging in ascending order, we get: 24, 34, 43, 50, 67, 78.

Here, n (no. of observations) = 6

$$6/2 = 3$$

Using the median formula, $\text{Median} = \frac{3\text{rd observation} + 4\text{th observation}}{2}$

$$= \frac{43 + 50}{2}$$

Median = 46.5

Median of Grouped Data

When the data is continuous and in the form of a frequency distribution, the median is found as shown below:

Step 1: Find the median class.

Let n = total number of observations i.e. $\sum f_i$

Note: Median class is the class where $\left(\frac{n}{2} \right)$ lies in cumulative frequency.

Step 2: Use the following formula to find the median.

$$\text{Median} = L_0 + \left(\frac{n}{2} - F_c \right) \times \frac{h}{f_0}$$

Where,

L_0 = Lower limit of median class

n = Total number of observations

F_c = Cumulative frequency of the class preceding the median class

f_0 = Frequency of the median class

h = Class size/Class width

Let's consider the following example to understand this better.

Example: Find the median marks for the following distribution:

Classes	0-10	10-20	20-30	30-40	40-50
Frequency	2	12	22	8	6

Solution: We need to calculate the cumulative frequencies to find the median.

Classes	Number of students	Cumulative frequency
0-10	2	2
10-20	12	14
20-30	22	36
30-40	8	44
40-50	6	50

$$n = 50$$

$$\frac{n}{2} = \frac{50}{2} = 25$$

$$\text{Median Class} = (20 - 30)$$

$$\text{Lower limit of median class, } L_0 = 20$$

$$\text{Total number of observations, } n = 50$$

$$\text{Cumulative frequency of the class preceding the median class, } F_c = 14$$

$$\text{Frequency of the median class, } f_0 = 22$$

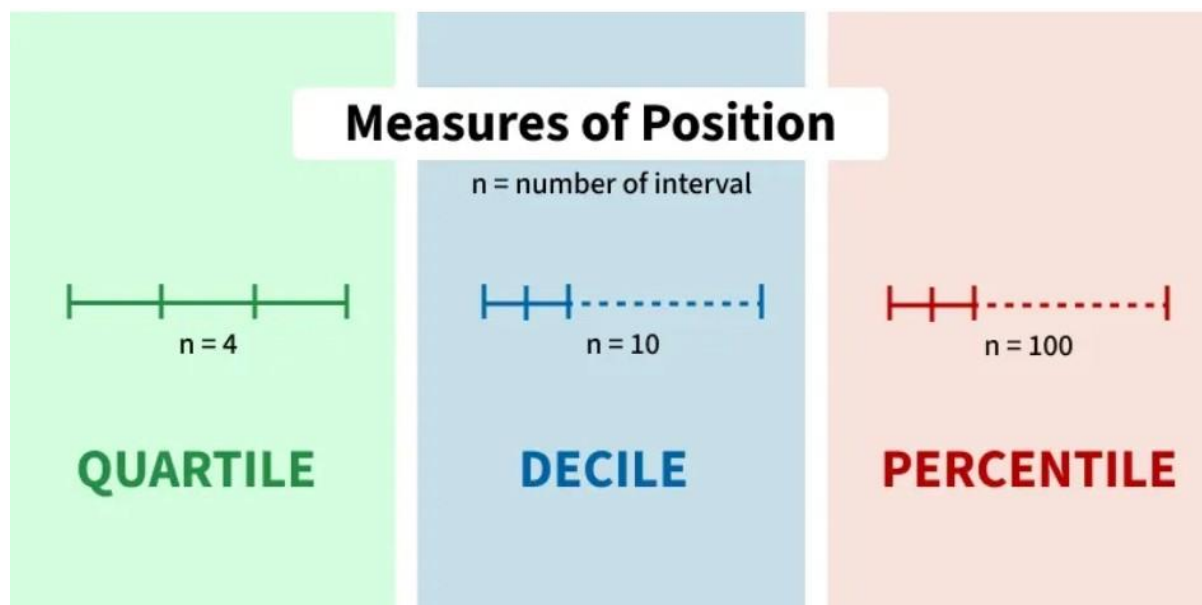
$$\text{Class size/Class width, } h = 10$$

Using Median formula:

$$\text{Median} = L_0 + \left(\frac{n}{2} - F_c \right) \times \frac{h}{f_0}$$

$$\begin{aligned}
 &= 20 + \left(\frac{50}{2} - 14\right) \times \frac{10}{22} \\
 &= 20 + (11) \times \frac{10}{22} \\
 &= 20 + 5 \\
 &= 25
 \end{aligned}$$

$\therefore \text{Median} = 25$



Quartiles

A quartile is a division of observations into four defined intervals based on the values of the observations and how they compare to the rest of the set of observations.

That sounds a lot more complicated than it is. To put it simply, the quartiles mark off 25% of the data, and so the quartiles are often named for the percentages:

1. The first quartile Q_1 , is the 25th percentile;
2. The second quartile Q_2 , is the 50th percentile; and
3. The third quartile Q_3 , is the 75th percentile.

N.B.: The second quartile is always the median of the data set.

Formula for quartile: $Q_i = L_i + \left(\frac{\frac{n}{4} - F_c}{f_i}\right) \times h$; $i = 1, 2, 3$

Where,

L_i = Lower limit of quartile class

n = Total number of observations

F_c = Cumulative frequency of the class preceding the quartile class

f_i = Frequency of the quartile class

h = Class size/Class width

Deciles

The deciles involve dividing a dataset into ten equal parts based on numerical values, each containing 10% of the data.

Formula for deciles: $D_i = L_i + \left(\frac{in}{10} - F_c\right) \times \frac{h}{f_i}; i = 1, 2, 3, \dots, 9$

Where,

L_i = Lower limit of decile class

n = Total number of observations

F_c = Cumulative frequency of the class preceding the decile class

f_i = Frequency of the decile class

h = Class size/Class width

Percentiles

Centiles are another term for *percentiles*. Percentiles divide a dataset into 100 equal parts, with each percentile representing the value below which a certain percentage of the data falls. These percentiles are commonly denoted as $P_1, P_2, P_3, \dots, P_{99}$.

Formula for percentiles: $P_i = L_i + \left(\frac{in}{100} - F_c\right) \times \frac{h}{f_i}; i = 1, 2, 3, \dots, 99$

Where,

L_i = Lower limit of percentile class

n = Total number of observations

F_c = Cumulative frequency of the class preceding the percentile class

f_i = Frequency of the percentile class

h = Class size/Class width

Exercise: number of births to women by current age in Natore-1994

Age in years	14.5-19.5	19.5-24.5	24.5-29.5	29.5-34.5	34.5-39.5	39.5-44.5	44.5-49.5
Number of births	677	1908	1737	1040	294	91	16

Evaluate Median, 3rd Quartile, 4th Decile and P_{85} .

Solution:

Make a calculation table from the data set:

Age in years	Number of births(f_i)	Cumulative Frequency
14.5-19.5	677	677
19.5-24.5	1908	2585
24.5-29.5	1737	4322
29.5-34.5	1040	5362
34.5-39.5	294	5656
39.5-44.5	91	5747
44.5-49.5	16	5763
Total	5763	

For median:

$$n = 5763$$

$$\frac{n}{2} = \frac{5763}{2} = 2881.5$$

$$\text{Median Class} = 24.5 - 29.5$$

$$\text{Lower limit of median class, } L_0 = 24.5$$

$$\text{Total number of observations, } n = 5763$$

$$\text{Cumulative frequency of the class preceding the median class, } F_c = 2585$$

$$\text{Frequency of the median class, } f_0 = 1737$$

$$\text{Class size/Class width, } h = 5$$

Using Median formula:

$$\begin{aligned} \text{Median} &= L_0 + \left(\frac{n}{2} - F_c \right) \times \frac{h}{f_0} \\ &= 24.5 + \left(\frac{5763}{2} - 2585 \right) \times \frac{5}{1737} \\ &= 24.5 + (296.5) \times \frac{5}{1737} \\ &= 24.5 + 0.85 \\ &= 25.35 \end{aligned}$$

$$\therefore \text{Median} = 25.35$$

For 3rd quartile:

$$n = 5763$$

$$\frac{3n}{4} = \frac{3 \times 5763}{4} = 4322.25$$

$$\text{Quartile Class} = 29.5 - 34.5$$

$$\text{Lower limit of quartile class, } L_i = 29.5$$

$$\text{Total number of observations, } n = 5763$$

$$\text{Cumulative frequency of the class preceding the quartile class, } F_c = 4322$$

$$\text{Frequency of the quartile class, } f_i = 1040$$

$$\text{Class size/Class width, } h = 5$$

Using quartile formula:

$$\begin{aligned} Q_3 &= L_i + \left(\frac{n}{2} - F_c \right) \times \frac{h}{f_i} \\ &= 29.5 + (4322.25 - 4322) \times \frac{5}{1040} \\ &= 29.5 + (0.25) \times \frac{5}{1040} \\ &= 29.5 + 0.001 \\ &= 29.501 \end{aligned}$$

$$\therefore Q_3 = 29.501$$

For 4th decile:

$$n = 5763$$

$$\frac{3n}{10} = \frac{4 \times 5763}{10} = 2305.2$$

decile Class = 19.5 – 24.5

Lower limit of decile class, $L_i = 19.5$

Total number of observations, $n = 5763$

Cumulative frequency of the class preceding the decile class, $F_c = 677$

Frequency of the decile class, $f_i = 1908$

Class size/Class width, $h = 5$

Using decile formula:

$$\begin{aligned} D_4 &= L_i + \left(\frac{4n}{10} - F_c \right) \times \frac{h}{f_i} \\ &= 19.5 + (2305.2 - 677) \times \frac{5}{1908} \\ &= 19.5 + (1628.2) \times \frac{5}{1908} \\ &= 19.5 + 4.27 \\ &= 23.77 \end{aligned}$$

$$\therefore D_4 = 23.77$$

For 85% percentile:

$$n = 5763$$

$$\frac{85n}{100} = \frac{85 \times 5763}{100} = 4898.55$$

Percentile Class = 29.5 – 34.5

Lower limit of percentile class, $L_i = 29.5$

Total number of observations, $n = 5763$

Cumulative frequency of the class preceding the percentile class, $F_c = 4322$

Frequency of the percentile class, $f_i = 1040$

Class size/Class width, $h = 5$

Using percentile formula:

$$\begin{aligned} P_{85} &= L_i + \left(\frac{85n}{100} - F_c \right) \times \frac{h}{f_i} \\ &= 29.5 + (4898.55 - 4322) \times \frac{5}{1040} \\ &= 29.5 + (576.55) \times \frac{5}{1040} \\ &= 29.5 + 2.77 \\ &= 32.27 \end{aligned}$$

$$\therefore P_{85} = 32.27$$

Mode

The value which appears most often in the given data i.e. the observation with the highest frequency is called a mode of data.

Ungrouped Data

For ungrouped data, we just need to identify the observation which occurs maximum times.

Mode = Observation with maximum frequency

For example in the data: 6, 8, 9, 3, 4, 6, 7, 6, 3, the value 6 appears the most number of times. Thus, mode = 6. An easy way to remember mode is: Most Often Data Entered. Note: A data may have no mode, 1 mode, or more than 1 mode. Depending upon the number of modes the data has, it can be called unimodal, bimodal, trimodal, or multimodal.

The example discussed above has only 1 mode, so it is unimodal.

Grouped Data

When the data is continuous, the mode can be found using the following steps:

Step 1: Find modal class i.e. the class with maximum frequency.

Step 2: Find mode using the following formula:

$$\text{Mode} = L_0 + \left(\frac{f_m - f_1}{2f_m - f_1 - f_2} \right) \times h$$

Where,

L_0 = Lower limit of modal class,

f_m = Frequency of modal class,

f_1 = Frequency of class preceding modal class,

f_2 = Frequency of class succeeding modal class,

h = Class width

Mode formula equivalently is written as follows as well:

$$\text{Mode} = L_0 + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2} \right) \times h$$

Where,

L_0 = Lower limit of modal class,

Δ_1 = Frequency of modal class- Frequency of class preceding modal class

Δ_2 = Frequency of modal class- Frequency of class succeeding modal class

h = Class width

Consider the following example to understand the formula.

Example: Find the mode of the given data:

Marks Obtained	0-20	20-40	40-60	60-80	80-100
Number of students	5	10	12	6	3

Solution:

Marks Obtained	0-20	20-40	40-60	60-80	80-100
Number of students	5	10	12	6	3

The highest frequency = 12, so the modal class is **40-60**.

L_0 = Lower limit of modal class = 40

f_m = Frequency of modal class = 12

f_1 = Frequency of class preceding modal class = 10

f_2 = Frequency of class succeeding modal class = 6

h = Class width = 20

Using the mode formula,

$$\begin{aligned} \text{Mode} &= L_0 + \left(\frac{f_m - f_1}{2f_m - f_1 - f_2} \right) \times h \\ &= 40 + \left(\frac{12 - 10}{2 \times 12 - 10 - 6} \right) \times 20 \\ &= 40 + \left(\frac{2}{8} \right) \times 20 \\ &= 40 + 5 \\ &= 45 \end{aligned}$$

$$\therefore \text{Mode} = 45$$

Example 1: If the mean of the following data is 20.6, find the missing frequency (p).

x	10	15	20	25	35
f	3	10	p	7	5

Solution: Let us make the calculation table for this:

x_i	f_i	$x_i f_i$
10	3	30
15	10	150
20	p	20p
25	7	175
35	5	175
Total	$\sum f_i = 25 + p$	$\sum x_i f_i = 530 + 20p$

$$\begin{aligned} \text{Mean} &= \frac{\sum x_i f_i}{\sum f_i} \\ &= \frac{530 + 20p}{25 + p} \end{aligned}$$

$$\text{ATQ, } \frac{530 + 20p}{25 + p} = 20.6$$

$$\Rightarrow 20.6p + 515 = 20p + 530$$

$$\Rightarrow 20.6p - 20p = 530 - 515$$

$$\Rightarrow 0.6p = 15$$

$$\therefore p = 25$$

Example 2: The mean of 5 numbers is 18. If one number is excluded, their mean is 16. Find the excluded number.

Solution:

Given, $n = 5$, $\bar{x} = 18$

$$\bar{x} = \frac{\sum x_i}{n}$$

$$\Rightarrow \sum x_i = 5 \times 18 = 90$$

Thus, the total of 5 numbers = 90

Let the excluded number be "a".

Therefore, total of 4 numbers = $90 - a$

$$\text{Mean of 4 numbers} = \frac{90 - a}{4}$$

$$\Rightarrow 16 = \frac{90 - a}{4}$$

$$\Rightarrow 90 - a = 64$$

$$\therefore a = 26$$

Answer: The missing number is 26.

Example 3: A survey on the heights (in cm) of 50 girls of class X was conducted at a school and the following data were obtained:

Height (in cm)	120-130	130-140	140-150	150-160	160-170	Total
Number of girls	2	8	12	20	8	50

Calculate the mode and median of the above data.

Solution:

The highest frequency = 20, so the modal class is 150-160.

L_0 = Lower limit of modal class = 150

f_m = Frequency of modal class = 20

f_1 = Frequency of class preceding modal class = 12

f_2 = Frequency of class succeeding modal class = 8

h = Class width = 10

Using the mode formula,

$$\begin{aligned} \text{Mode} &= L_0 + \left(\frac{f_m - f_1}{2f_m - f_1 - f_2} \right) \times h \\ &= 150 + \left(\frac{20 - 12}{2 \times 20 - 12 - 8} \right) \times 10 \\ &= 150 + \left(\frac{8}{20} \right) \times 10 \\ &= 150 + 4 \\ &= 154 \end{aligned}$$

$$\therefore \text{Mode} = 154$$

To find the median, we need cumulative frequencies.

Consider the table:

Class Intervals	No. of girls (f_i)	Cumulative frequency
120-130	2	2
130-140	8	10
140-150	12	22
150-160	20	42
160-170	8	50

$$n = 50$$

$$\frac{n}{2} = \frac{50}{2} = 25$$

$$\text{Median Class} = (150 - 160)$$

$$\text{Lower limit of median class, } L_0 = 150$$

$$\text{Total number of observations, } n = 50$$

$$\text{Cumulative frequency of the class preceding the median class, } F_c = 22$$

$$\text{Frequency of the median class, } f_0 = 20$$

$$\text{Class size/Class width, } h = 10$$

Using Median formula:

$$\begin{aligned}
 \text{Median} &= L_0 + \left(\frac{n}{2} - F_c \right) \times \frac{h}{f_0} \\
 &= 150 + \left(\frac{50}{2} - 22 \right) \times \frac{10}{20} \\
 &= 150 + (3) \times \frac{1}{2} \\
 &= 150 + 1.5 \\
 &= 151.5
 \end{aligned}$$

$$\therefore \text{Median} = 151.5$$

Differences between Mean, Median, and Mode

Now, we have got enough knowledge about mean mode and median. Let us summarize the differences between mean median & mode in the table below.

Property	Mean	Median	Mode
Definition	Average value of a set of numbers.	Middle value of a set of numbers when they are arranged in order.	The most frequently occurring value in a set of numbers.
Calculation	Add up all the numbers and divide by the total number of numbers.	Arrange all the numbers in order, and the median is the middle number. If there are an even number of values, take the average of the two middle values.	Identify the value that appears most often in the set.
Affected by Outliers?	Yes	No	No